exploratory goal for the human species. If we're talking about spending this much money, we must justify the expense.

There are now other matters—clear, crying national needs—that cannot be addressed without major expenditures; at the same time, the discretionary federal budget has become painfully constrained. Disposal of chemical and radioactive poisons, energy efficiency, alternatives to fossil fuels, declining rates of technological innovation, the collapsing urban infrastructure, the AIDS epidemic, a witches' brew of cancers, homelessness, malnutrition, infant mortality, education, jobs, health care—there is a painfully long list. Ignoring them will endanger the well-being of the nation. A similar dilemma faces all the spacefaring nations.

Nearly every one of these matters could cost hundreds of billions of dollars or more to address. Fixing infrastructure will cost several trillion dollars. Alternatives to the fossil-fuel economy clearly represent a multitrillion-dollar investment worldwide, if we can do it. These projects, we are sometimes told, are beyond our ability to pay. How then can we afford to go to Mars?

If there were 20 percent more discretionary funds in the U.S. federal budget (or the budgets of the other spacefaring nations), I probably would not feel so conflicted about advocating sending humans to Mars. If there were 20 percent less, I don't think the most diehard space enthusiast would be urging such a mission. Surely there is some point at which the national economy is in such dire straits that sending people to Mars is unconscionable. The question is where we draw the line. Plainly such a line exists, and every participant in these debates should stipulate where that line should be drawn, what fraction of the gross national product for space is too much. I'd like the same thing done for "defense."

Public opinion polls show that many Americans think the NASA budget is about equal to the defense budget. In fact, the entire NASA budget, including human and robotic missions and aeronautics, is about 5 percent of the U.S. defense budget. How much spending for defense actually weakens the country? And even if NASA were cancelled altogether, would we free up what is needed to solve our national problems?